Supply Chain Management
Unit 1

Introduction to Operations Management

History of operations management

Types of manufacturing systems

Roles and responsibilities of operations manager,

Product operations and service operations,

Current trends in operations management

Operations Management

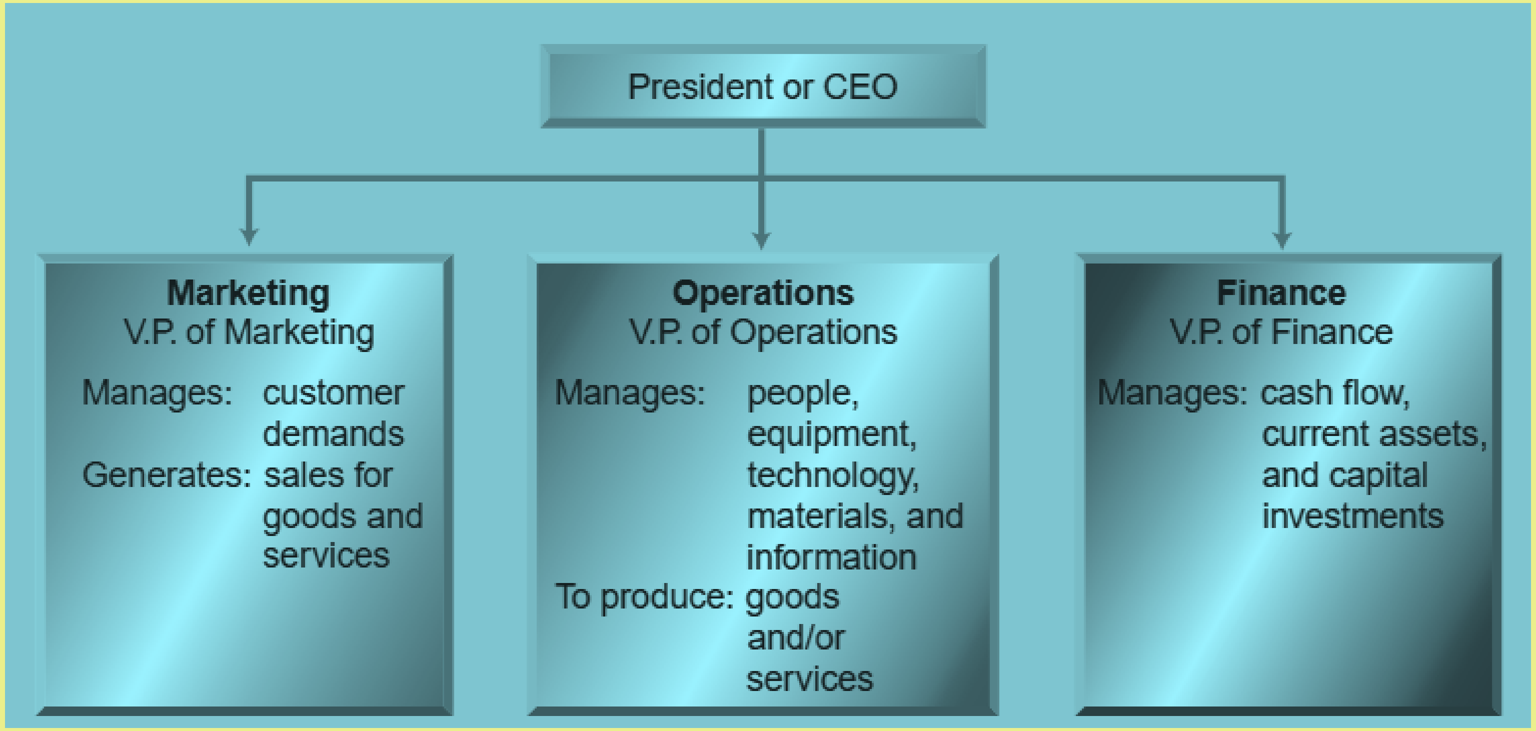
- Several authors emphasize Operations Management:
- **APICS Dictionary (1995)**
  - *A field of study that focuses on the planning, scheduling, use, and control of a manufacturing or service organization, through the study of design engineering, industrial engineering, MIS, quality management, production management, accounting, and other functions as they affect operations.*
- **Barnett (1996)**
  - *Operations Management is concerned with the efficient conversion of an organization's resources into the goods and services that it has been set up to provide.*
- **Naylor (1996)**
  - *Operations Management deals with creating, operating, and controlling a transformation system that converts various inputs into goods and services required by customers.*
- **Waters (1991)**
  - *Operations Management is concerned with all activities involved in producing a product or providing a service; it is responsible for transforming inputs into useful outputs.*
- **Stevenson (1993)**
  - *Operations Management is the management of systems or processes that create goods and/or provide services.*

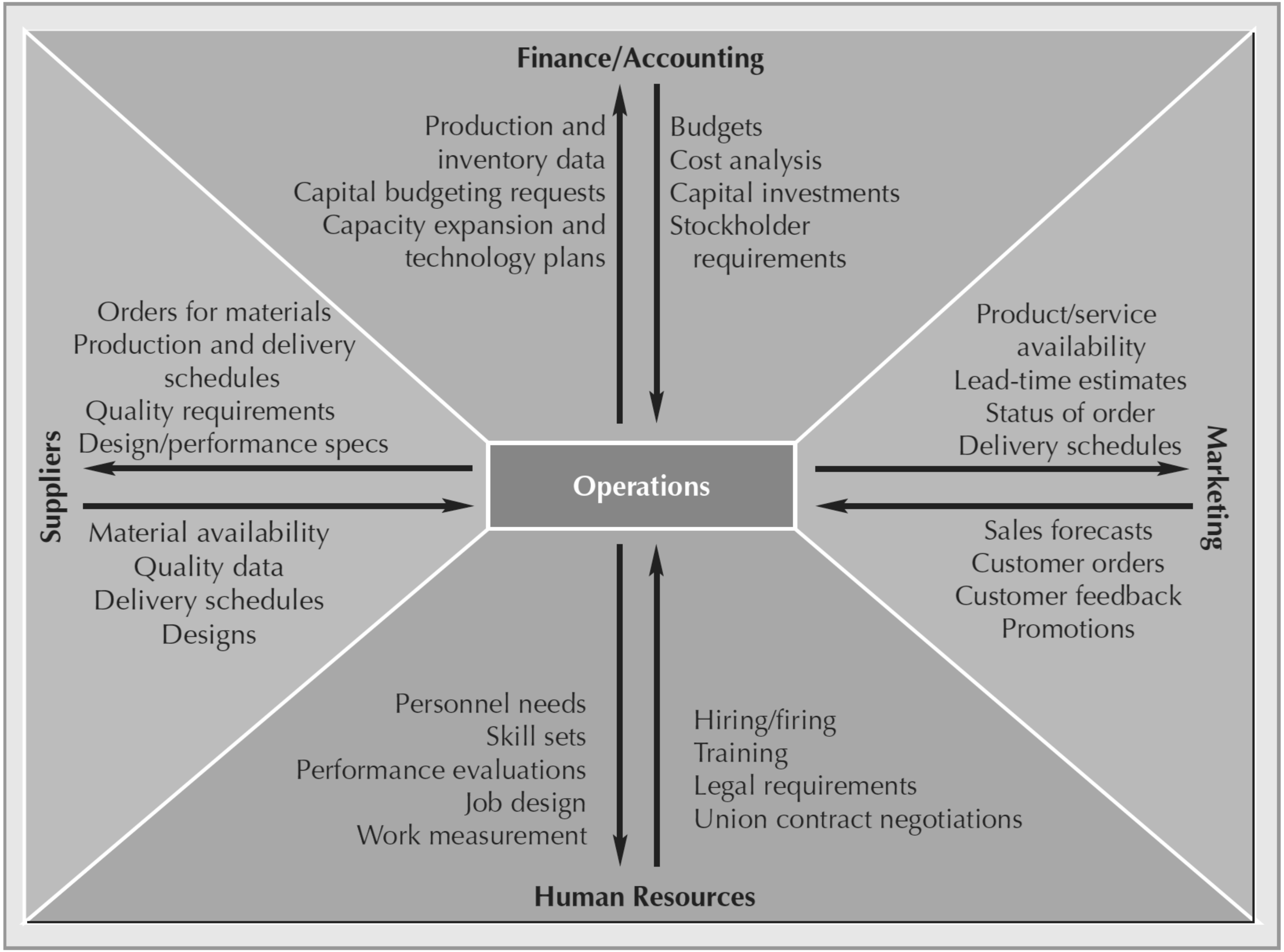
Cont..

- **Production and Operations Management (POM)** is concerned with the application of basic management concepts, principles, and practices to those areas of an organization that are directly related to the **production of goods and/or services**.
- It involves converting inputs into outputs using physical resources in a manner that provides the desired utility to customers while meeting organizational objectives, such as **efficiency, effectiveness, and adaptability**.

Major Business Functions

- Every business has **three primary functions**:
  - **Finance:** manages cash flow, assets, and investments.
  - **Marketing:** generates sales, understands customer needs, and drives demand.
  - **Operations Management:** plans, organizes, coordinates, and controls resources to **produce goods and services**.
- Other functions, such as accounting, HR, purchasing, and engineering, **support these three core functions**.
- **Operations is that part of a business organisation that is responsible for producing goods and/or services.**




Finance/Accounting
Production and inventory data
Capital budgeting requests
Capacity expansion and technology plans
Budgets
Cost analysis
Capital investments
Stockholder requirements
Orders for materials
Production and delivery schedules
Quality requirements
Design/performance specs
Suppliers
Material availability
Quality data
Delivery schedules
Designs
Operations
Product/service availability
Lead-time estimates
Status of order
Delivery schedules
Marketing
Sales forecasts
Customer orders
Customer feedback
Promotions
Personnel needs
Skill sets
Performance evaluations
Job design
Work measurement
Hiring/firing
Training
Legal requirements
Union contract negotiations
Human Resources

Understanding Operations Management (OM)

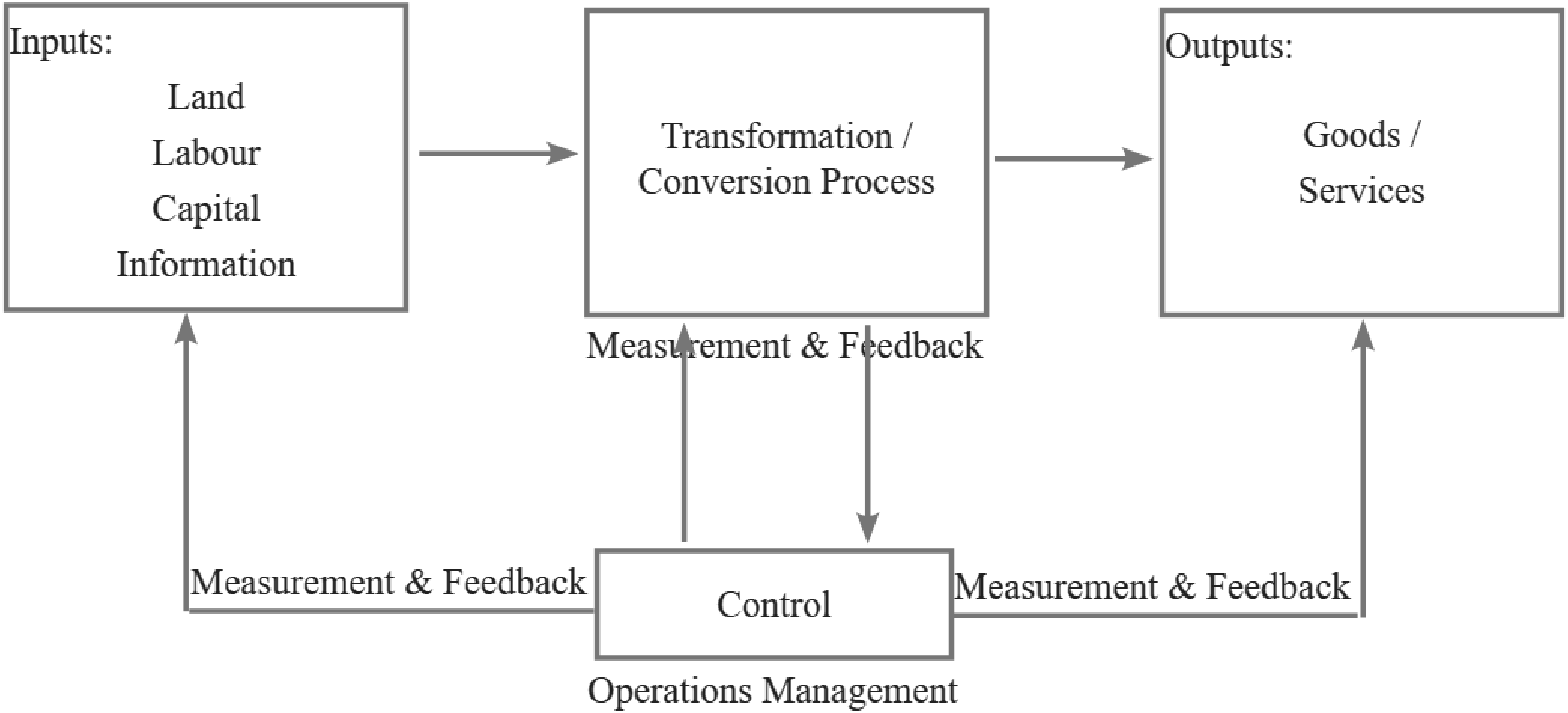
- Many people are already familiar with OM without realizing it; they may even use OM techniques in their daily work.
- OM is one of the **most critical business functions today**.
- Being involved in OM positions a company at the **frontier of business competition**.

- Companies face a **different competitive environment** than in the past.
- Survival requires focus on:
  - **Quality**
  - **Time-based competition**
  - **Efficiency**
  - **International perspectives**
  - **Customer relationships**
- Factors driving this change:
  - **Global competition**
  - **E-business and the Internet**
  - **Technological advances**

Understanding Operations Management (OM)

- Companies must be **flexible, responsive, lean, and agile** to remain competitive.
  - **Companies Excelling Through OM**
  - Examples: **Wal-Mart, Southwest Airlines, General Electric, Starbucks, Apple, Toyota, FedEx, Procter & Gamble**
  - World-class performance is often achieved through a **strong operations management focus**.
- **Purpose of Studying OM**
  - Helps you understand how to give your organization a **competitive advantage**.
  - Applies across all business areas: **marketing, finance, MIS, and operations**.
  - Teaches how to deliver goods/services **cheaper, better, and faster**.
  - OM concepts **impact all organizational functions** and everyday life.

Operation Function involves the conversation of Inputs and Outputs



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- **Inputs:** may be in the form of materials, information and knowledge. Transforming resources include human resources, facilities like buildings, machinery, and equipment – these are sometimes referred to as ‘capital’.

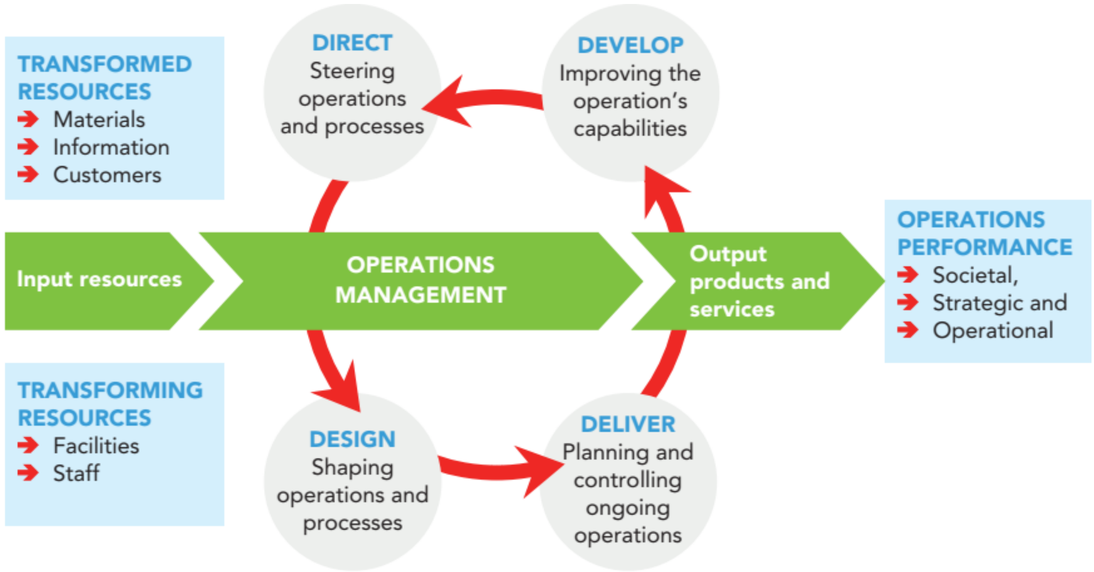
- **Outputs:** The transformation processes produce both goods and services.
  - The goods and services that are finally produced are the output.
  - In the process of transformation, it is also important for the operations management function to ensure minimization of the environmental impact of waste over the life cycle of their products, up to the point of final disposal, protecting the health and safety of employees and of the local community and adopting ethical behaviour in relation to the social impact of transformation processes, both locally and globally.

Transformation processes

- The **transformation process** refers to the method by which **inputs** are converted into **outputs** that provide value and satisfy customer needs. Inputs may include **materials, information, and knowledge**, while transforming resources typically include **human resources**, **facilities**, **equipment**, and **capital**.

- **Types of Transformations**
- When inputs are raw materials, transformation processes may involve:
  - **Physical changes** to materials or customers
  - **Changes in the location** of materials or information
  - **Changes in ownership** of goods
  - **Storage or accommodation** of materials
  - **Processing of information** to meet customer expectations
  - **Changes in the physiological or psychological state of customers**

System	Primary Inputs	Resources	Primary Transformation Functions	Typical Desired Output
Hospital	Patients	MDs, nurses, medical supplies, equipment	Health care	Healthy individuals
Restaurant	Hungry Customer	Food, chef, wait staff, environment	Well-prepared, well-served food; agreeable environment	Satisfied customers
Automobile factory	Sheet steel, engine parts	Tools, equipment, workers	Fabrication and assembly of cars	High-quality cars
Collage or university	High school graduates	Teachers, books, Classrooms	Teaching knowledge and skills	Educated individuals
Department store	Shoppers	Displays, stocks of goods, sales clerks	Attract shoppers, promote products, fill orders	Sales to satisfied customers
Distribution Center	Stock keeping units (SKUs)	Storage bins, stock pickers	Storage and redistribution	Fast delivery, availability of SKUs



- **Operations Management**:
  - This central component is responsible for managing the entire process of converting input resources into output products and services. It ensures efficient and effective utilization of resources.
- **Operations Activities**:
  - **Design**: Shaping operations and processes. This involves planning and organizing how resources are to be used and how processes should be structured.
  - **Deliver**: Planning and controlling ongoing operations. This step focuses on ensuring that operations run smoothly and deliver the expected output.
  - **Develop**: Improving the operation's capabilities. This involves continuous improvement and enhancement of processes and resources to increase efficiency and effectiveness.
  - **Direct**: Steering operations and processes. This activity involves guiding and managing the operations to align with the organization's goals and objectives.

Objectives of Operations Management

- The objectives of Operations Management (OM) can be broadly classified into two categories:
  - Customer Service
  - Resource Utilisation
- (i) Customer Service
  - The **primary objective of operations management** is to provide effective **customer service**, which means satisfying customer needs and expectations.
  - Customer service is achieved when operations management delivers products or services that meet customer requirements in terms of:
    - **Specification**
    - **Cost**
    - **Timing**
    - Thus, the fundamental objective of OM is to provide **"the right product or service, at the right price, at the right time."**
- These three aspects—**specification, cost, and timing**—are the principal sources of customer satisfaction. Therefore, they form the key dimensions of the customer service objective in operations management.

Aspects of Customer Service

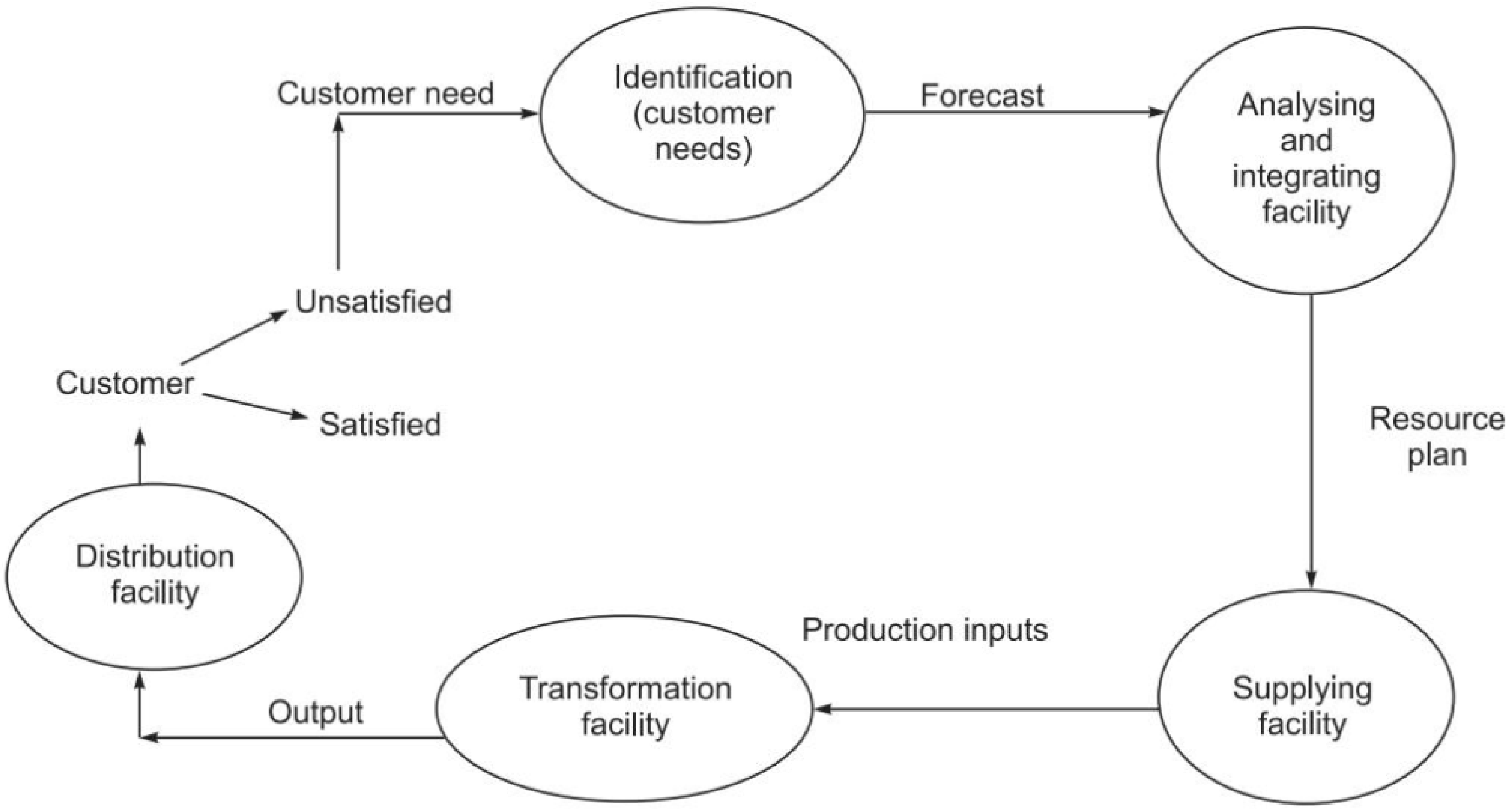
Principal Function	Principal Customer Wants	Cost Consideration	Timing Consideration
Manufacturing	Goods of the requested or acceptable specification	Purchase price or cost of obtaining goods	Delivery time from order to receipt
Transport	Movement of goods of acceptable specification	Cost of movement	(i) Time taken for movement(ii) Waiting or delay before movement starts
Supply	Goods of requested or acceptable specification	Purchase cost or cost of supply	Delivery delay from order/request to receipt
Service	Treatment or service of acceptable specification	Cost of treatment	(i) Duration of service(ii) Waiting time before service begins

- **(ii) Resource Utilization**
  - The **second major objective of Operations Management** is **to effectively utilize resources** while satisfying customer needs. Customer service must be achieved **through efficient operations**, ensuring optimal use of available resources.
  - Inefficient use of resources or failure to provide adequate customer service can lead to the **commercial failure** of a business.
- Operations management is primarily concerned with:
  - **Obtaining maximum output from available resources**
  - **Minimizing loss, underutilization, and waste**
  - **Measures of Resource Utilization**
  - The extent of resource utilization can be expressed through:
  - **Time utilization** (proportion of available time used)
  - **Space utilization**
  - **Activity or capacity utilization levels**
  - These measures indicate how effectively the **potential or capacity of resources** is being used. This constitutes the **resource utilization objective** of operations management.

Steps in Operations Management

- **Step 1: Identification Of Customer Needs**
  - Understanding Customer Requirements
  - Translating These Needs Into A **Clear Demand Forecast**
- **Step 2: Resource Planning**
  - Analysis Of Demand Forecasts
  - Planning And Integration Of Facilities And Resources
- **Step 3: Execution Of Resource Plans**
  - Procuring Resources Through **Internal Or External Suppliers**
  - Generation Of Required Inputs
- **Step 4: Conversion / Transformation**
  - Movement Of Inputs Into The **Conversion Facility**
  - Transformation Of Inputs Into Outputs (Physical Goods Or Intangible Services)
- **Step 5: Delivery To Customers**
  - Distribution Of Goods Or Services
  - Ensuring **Customer Satisfaction** Through Efficient Delivery Systems.
  - **Conclusion:** Operations Management Plays a Central Role in Organizations by Managing Processes That Convert Inputs into Outputs Efficiently and Effectively. It Ensures That Products And Services Are **Designed, Produced, And Delivered** In Line With Customer Expectations And Organizational Objectives.

Sequential Steps for Customer Satisfaction



Characteristics of Operations Management

- The important characteristics of Operations Management (OM) are as follows:

- **1. Basic Business Function:** Operations management is the **core function of any business organization**. It is a **line function** that interacts with all other functional areas, such as marketing, finance, and human resources. OM is responsible for producing **quality goods or services**, often described as a **package of attributes**, to satisfy customer needs, which is the ultimate objective of any business.

- **2. Responsibility to Produce and Deliver on Time:** Operations management deals with the **activities, decisions, and responsibilities** of production or operations managers. These managers are responsible for ensuring the **efficient production and timely delivery** of goods and services to customers.

- **3. Real Value Addition Function:** Operations management is the function that **adds real value** by applying the **transformation model**. Through this model, inputs are converted into outputs (goods or services), ensuring value addition that satisfies customer requirements.

- **4. Changing Properties to Gain Competitive Advantage**
  - Operations management changes:
  - **Physical properties** (e.g., raw materials into finished goods)
  - **Informational properties**
  - **Possession and location**
  - **Storage or accommodation**
  - **Physiological state** (e.g., healthcare services)
  - These changes help organizations gain a **competitive edge** over rivals by delivering superior value.

- **5. Decision-Making as a Core Function**
  - Decision-making is central to operations management and includes:
  - **Strategic (long-term) decisions**
  - **Tactical or intermediate decisions**
  - **Operational (short-term) decisions**
  - These decisions affect capacity, process design, scheduling, quality, and cost.

- **6. Use of Management Models**
  - Operations managers use various **management models and approaches** to manage operations effectively, including:
  - **Classical models**
  - **Behavioural models**
  - **Systems and quantitative models**
  - These approaches help in efficiently **converting inputs into desired outputs**.

Components of Operations Management

- Operations Management consists of several interrelated components that collectively ensure the **efficient design, production, and delivery** of goods and services.
- **1. Product Development**
- Product development is the process of **designing, creating, and marketing a product or service**.
  A product may be:
  - **New to the market**, or
  - **New to the producing organization**, or
  - An **improved version** of an existing product through modifications or enhancements.
- Product development ensures that customer needs are effectively translated into **marketable and competitive offerings**.

- **2. Process Design and Management**
- Process design involves deciding:
- Which processes should be performed in-house, and which should be outsourced?
- This component focuses on the **design and management of physical and service processes** used to produce goods and services.
- Key decisions include:
  - Selection of production methods
  - Layout of facilities
  - Technology and equipment choices
- Effective process design ensures **efficiency, quality, and cost control**.

- **3. Supply Chain Management:** Supply chain management involves the **network of organizations, activities, people, technology, information, and resources** required to move a product or service from the producer to the **final customer**.
- Supply chain activities include:
  - Procurement of raw materials
  - Production and processing
  - Warehousing and distribution
  - Delivery to end-users
  - Efficient supply chain management ensures the **timely availability of products**, reduced costs, and improved customer satisfaction.

- **Conclusion:** Product development, process design, and supply chain management are **core components of operations management**. Together, they help organizations achieve **operational efficiency, competitiveness, and customer satisfaction**.

Operations Management: A Framework

- Operations Management can be understood through a **conceptual framework** that describes the key activities performed by an operations manager. This framework integrates **People, Product, Process, Plant, and Programme,** which together ensure effective and efficient operations.

- **1. People:** People are the **most valuable assets** of an organization. Effective operations management requires:
  - Proper **utilization of human resources**
  - A good **match between people and their jobs**
  - Consideration of skills, knowledge, experience, and motivation
  - Efficient use of people improves productivity, quality, and overall organizational performance.

- **2. Product:** The **product** acts as the **interface between marketing and production**. Therefore, coordination among all business functions is essential.
- Agreement must be reached on:
  - Quality
  - Durability and reliability
  - Quantity and performance
  - Selling price
  - Delivery date
- While making product-related decisions, the following factors must be considered:
  - Market needs
  - Organizational culture
  - Legal and regulatory constraints
  - Environmental and social demands
- Operations alone cannot decide these aspects; **inter-functional coordination** is essential.

- **3. Process**
  - The **process** defines *how* a product is made or a service is delivered.
- Key considerations include:
  - Selecting the most suitable method of production or service delivery
  - Improving or discovering new and better ways of performing operations
  - Aligning processes with the **knowledge, skills, and intellect** of the workforce
  - An effective process ensures **cost efficiency, quality, and flexibility**.

- **4. Plant:** The **plant** includes physical facilities such as buildings, machines, and equipment. These assets must match:
    - Product requirements
    - Market demand
    - Operator capabilities
    - Organizational objectives
  - The **plant layout** depends on:
    - Type of production
    - Volume of demand
    - Nature and variety of products
    - Proper plant planning enhances workflow and reduces waste.

- **5. Programme:** A **programme** refers to the **step-by-step plan or timetable of production**.
- It involves:
  - Scheduling activities
  - Coordinating resources
  - Meeting multiple objectives simultaneously, such as cost, quality, and delivery time
  - Production programmes can be complex and require careful planning and control.

Significance of Production and Operations Management in Business

- Production and Operations Management (POM) plays a **crucial role in business**, as it is the **core functional area** directly associated with creating value for customers.

- **1. Core Line Function of Business:** Operations management is a **line function** and accounts for **more than 70% of manpower deployment and capital investment** in most organizations. It directly contributes to the creation of goods and services and therefore forms the backbone of business operations.

- **2. Supports Corporate Strategy:** The operations management function plays a vital role in **supporting and implementing corporate strategy**. Strategic decisions related to:
  - Capacity
  - Technology
  - Process design
  - Location and layout are translated into action through operations management.

- **3. Information Exchange Hub:** Operations management acts as a **central information exchange point** by sharing critical information related to:
  - Product design
  - Process selection
  - Quality standards
  - Supply chain management
  - Logistics and delivery time
  - Technology development
  - This coordination ensures smooth functioning across departments.

- **4. Value Creation for Customers:** The **value perceived by customers** in a product or service arises from a **complete package of attributes**, such as:
  - Quality
  - Performance
  - Reliability
  - Price
  - Availability
  - Delivery time
  - This value can be created **only through production and operations management**, which integrates these attributes into the final product or service offering.

- **5. Customer Satisfaction and Business Survival:** An organization survives and grows by **delivering what customers want**, namely:
  - **Right quality**
  - **Right quantity**
  - **Right time**
  - **Right place**
  - **Right price**
- Operations management ensures that products and services are produced and delivered in line with these requirements, thereby ensuring **customer satisfaction and competitiveness**.

Importance of Operations Management

- Operations Management (OM) is essential for every business organization because it deals with the **production of goods and delivery of services**, which are the core activities that create value. Effective OM ensures high productivity, quality, and customer satisfaction while controlling costs and improving competitiveness.

1. Core Function of Every Organization

- Operations is one of the two line functions (along with sales). All other departments—finance, marketing, HR, IT—exist to support operations. Without efficient operations, the organization cannot produce goods or deliver services effectively.

2. Ensures Efficient Use of Resources

- OM helps in the optimal utilization of:
- Materials
- Machines
- Human resources
- Information
- Capital
- Effective resource use reduces waste, lowers costs, and increases profitability.

- **3. Improves Productivity**
  - Productivity (output per unit input) is directly influenced by operations processes.
    Higher productivity leads to:
  - Lower production cost
  - Higher competitiveness
  - Better financial performance

- **4. Enhances Quality and Customer Satisfaction**
  - OM designs processes that ensure consistent product/service quality.
    Improved quality results in:
  - Higher customer satisfaction
  - Fewer complaints and defects
  - Better brand image

5. Facilitates Coordination with Other Business Functions

- – Operations interacts closely with:
- – **Finance**
  - • Budgeting
  - • Investment analysis
  - • Cash-flow planning
- **Marketing**
  - – Meeting customer demands
  - – Ensuring timely delivery
  - – Developing products customers want
- **Accounting**
  - – Cost control
  - – Performance evaluation
- **Information Systems**
  - – Automation
  - – Real-time data for decision-making
  - – Good coordination ensures smooth functioning across the organization.

6. Supports Strategic Decision-Making

- – Operations Management plays a key role in long-term decisions such as:
- – Capacity planning
- – Facility location and layout
- – Technology selection
- – Supply chain strategy
- – Product and service design
- – These decisions determine the organization’s future performance and competitiveness.

7. Enables Innovation and Continuous Improvement

- OM encourages practices such as:
- Lean manufacturing
- Total Quality Management (TQM)
- Six Sigma
- Automation and digitization
- These innovations help reduce waste, improve processes, and stay competitive.

8. Essential in Both Manufacturing and Services

- OM is critical not just for factories but also for:
- Banking
- Healthcare
- Hospitality
- IT services
- Education
- Transportation
- It improves service speed, reliability, and customer experience.

Scope of Production and Operations Management

Example of DELL

- Proper management of the operations function has led to success for many companies.
- For example, in 1994, Dell Computer Corporation was a second-rate computer maker that managed its operations similarly to others in the industry.
- Then Dell implemented a new business model that completely changed the role of its operations function.
- Dell has developed new and innovative ways of managing the operations function, which have become one of today’s best practices.
- These changes enabled Dell to deliver customized products to customers rapidly and at a lower cost.
- Today, Dell's customized computers can be delivered to the customer within 36 hours, at a price 10–15 percent lower than the industry standard.
- Dell’s model is one that many have tried to emulate, and it is the key to its status as a leader in the industry.

Example of Kozmo.com

- Kozmo.com promised delivery of a wide range of products—from videos to ice cream—**within one hour**.
- The company was **technology-driven** and initially experienced rapid success.
- However, this success led to **overly rapid expansion**, which the company’s operations could not support.
- Kozmo failed to manage critical operational aspects such as:
  - Inventory control
  - Delivery coordination
  - Cost management
- As a result, the company faced **excess inventory, poor delivery performance, and heavy financial losses**.
- Despite attempts to improve operations, the problems could not be corrected in time, and Kozmo.com **ceased operations in April 2001**.
- This example highlights that **making promises is easier than delivering them**, and effective operations management is essential to fulfill customer commitments.

Operations Management in the Web-Based sector

- The Web-based age has created a **highly competitive environment for online shopping**, presenting **unique challenges for operations management**.
- The Internet enables customers to purchase a wide range of products online, including CDs, books, groceries, prescription medicines, and even automobiles.
- While the Internet has provided customers with greater **flexibility and convenience**, it has also created one of the **biggest challenges for companies**—**delivering exactly what the customer ordered at the promised time**.
- As illustrated by the case of **Kozmo.com**, making promises on a website is relatively easy, but **fulfilling those promises through effective operations is much more complex**. Ensuring delivery from **“mouse to house”** is the responsibility of operations management and requires careful coordination of multiple activities.

Operational Challenges in E-Commerce

- Many dot-com companies in the 1990s failed because they could not manage these operational challenges profitably. To meet customer expectations, companies must:
- Accurately **forecast customer demand**
- Maintain **adequate inventory levels**
- Manage **distribution centers and warehouses**
- Operate and coordinate **transportation and delivery systems**
- Control costs while ensuring **customer satisfaction**
- **Role of Operations Strategy**
- Companies like **Amazon.com** manage most operational activities internally to ensure speed and reliability.
- Other firms **outsource logistics functions** such as inventory management and deliveries to specialized companies like **UPS**.
- Competition among e-tailers has intensified, with customers demanding:
  - Faster delivery times
  - Greater customization
  - **Same-day delivery**, especially in metropolitan areas
    - Example: **Barnesandnoble.com** offers same-day delivery in cities such as Manhattan, Los Angeles, and San Francisco.

Summery

- In the digital era, operations management has become critical to business survival and competitiveness.
- Effective management of the transformation process—from inputs to outputs—along with timely delivery and integration of customer feedback, determines whether online businesses succeed or fail.

Nature of goods or services

- Although both **manufacturing** and **service operations** fall under the scope of Production and Operations Management, they differ significantly in their characteristics. The key differences are as follows:
  - **Nature of output**
  - **Nature of the job**
  - **Nature of quality assurance**
  - **Nature of performance measurement**

Nature of Output

- **Manufacturing Operations**
  - Output is **tangible**, physical products.
  - Products can be **stored**, **transported**, and **inspected** before delivery.
  - Production and consumption occur at **different times**.
  - Examples: automobiles, furniture, machinery.
- **Service Operations**
  - Output is **intangible**, such as experiences or activities.
  - Services **cannot be stored** or inventoried.
  - Production and consumption occur **simultaneously**.
- Examples: healthcare, education, banking, transport services.

Nature of Job

- **Manufacturing**
  - Jobs are often **machine-oriented**, repetitive, and standardized.
  - High use of **automation and technology**.
  - Limited direct contact with customers during production.
- **Service**
  - Jobs involve **high customer interaction**.
  - Work is often **customized**, variable, and people-oriented.
  - Service providers must adapt to customer needs in real-time.

Nature of Quality Assurance

- **Manufacturing**
  – Quality can be measured through **specifications, tolerance limits, inspections**, and **testing**.
  – Defects can often be **found and corrected** before the product reaches the customer.
  – Emphasis on **conformance to standards**.
- **Service**
- Quality is based on **customer perception** and **experience**.
- Difficult to inspect before delivery since the service occurs instantly.
- Emphasis on **responsiveness, empathy, reliability, and service consistency**.

Nature of Performance Measurement

- **Manufacturing**
- Performance is measured using **quantitative metrics** such as:
  - Productivity
  - Output volume
  - Cost per unit
  - Machine efficiency
  - Defect rates
- **Service**
- Measurement includes **both quantitative and qualitative indicators**, such as:
  - Customer satisfaction
  - Waiting time
  - Service speed
  - Employee behavior
  - Service consistency
  - Customer feedback ratings

History of Operations Management

Operations Management (OM) has evolved over centuries alongside the development of production systems, technology, and management thought.

Era	Events/Concepts	Dates	Originator
Industrial Revolution	Steam engine	1769	James Watt
	Division of labor	1776	Adam Smith
	Interchangeable parts	1790	Eli Whitney
Scientific Management	Principles of scientific management	1911	Frederick W. Taylor
	Time and motion studies	1911	Frank and Lillian Gilbreth
	Activity scheduling chart	1912	Henry Gantt
	Moving assembly line	1913	Henry Ford

Era	Events/Concepts	Dates	Originator
Human Relations	Hawthorne studies	1930	Elton Mayo
	Motivation theories	1940s	Abraham Maslow
		1950s	Frederick Herzberg
		1960s	Douglas McGregor
Operations Research	Linear programming	1947	George Dantzig
	Digital computer	1951	Remington Rand
	Simulation, waiting line theory, decision theory, PERT/CPM	1950s	Operations research groups
	MRP, EDI, EFT, CIM	1960s, 1970s	Joseph Orlicky, IBM and others

Era	Events/Concepts	Dates	Originator
Quality Revolution	JIT (just-in-time)	1970s	Taiichi Ohno (Toyota)
	TQM (total quality management)	1980s	W. Edwards Deming, Joseph Juran
	Strategy and operations	1980s	Wickham Skinner, Robert Hayes
	Business process reengineering	1990s	Michael Hammer, James Champy
	Six Sigma	1990s	GE, Motorola

Era	Events/Concepts	Dates	Originator
Internet Revolution	Internet, WWW, ERP, supply chain management	1990s	ARPANET, Tim Berners-Lee SAP, i2 Technologies, ORACLE
	E-commerce	2000s	Amazon, Yahoo, eBay, Google, and others
Globalization	WTO, European Union, and other trade agreements, global supply chains, outsourcing, BPO, Services Science	1990s 2000s	Numerous countries and companies

Types of Manufacturing Systems

- A manufacturing system refers to the arrangement and operation of facilities, equipment, and processes used to produce goods from raw materials.
- Manufacturing systems are divided into two broad categories
  - **Intermittent Production** and
  - **Continuous Production.**

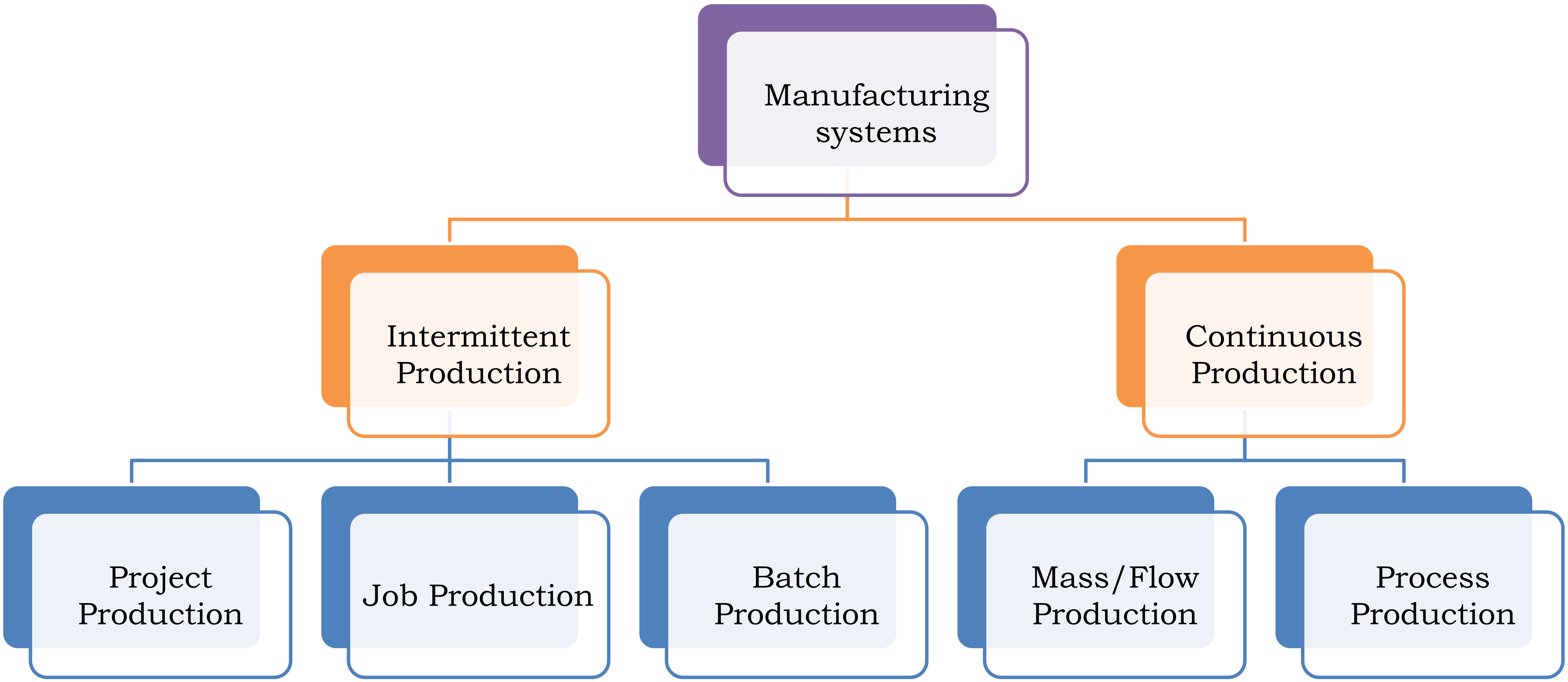
1. Intermittent Production (also called discrete production)

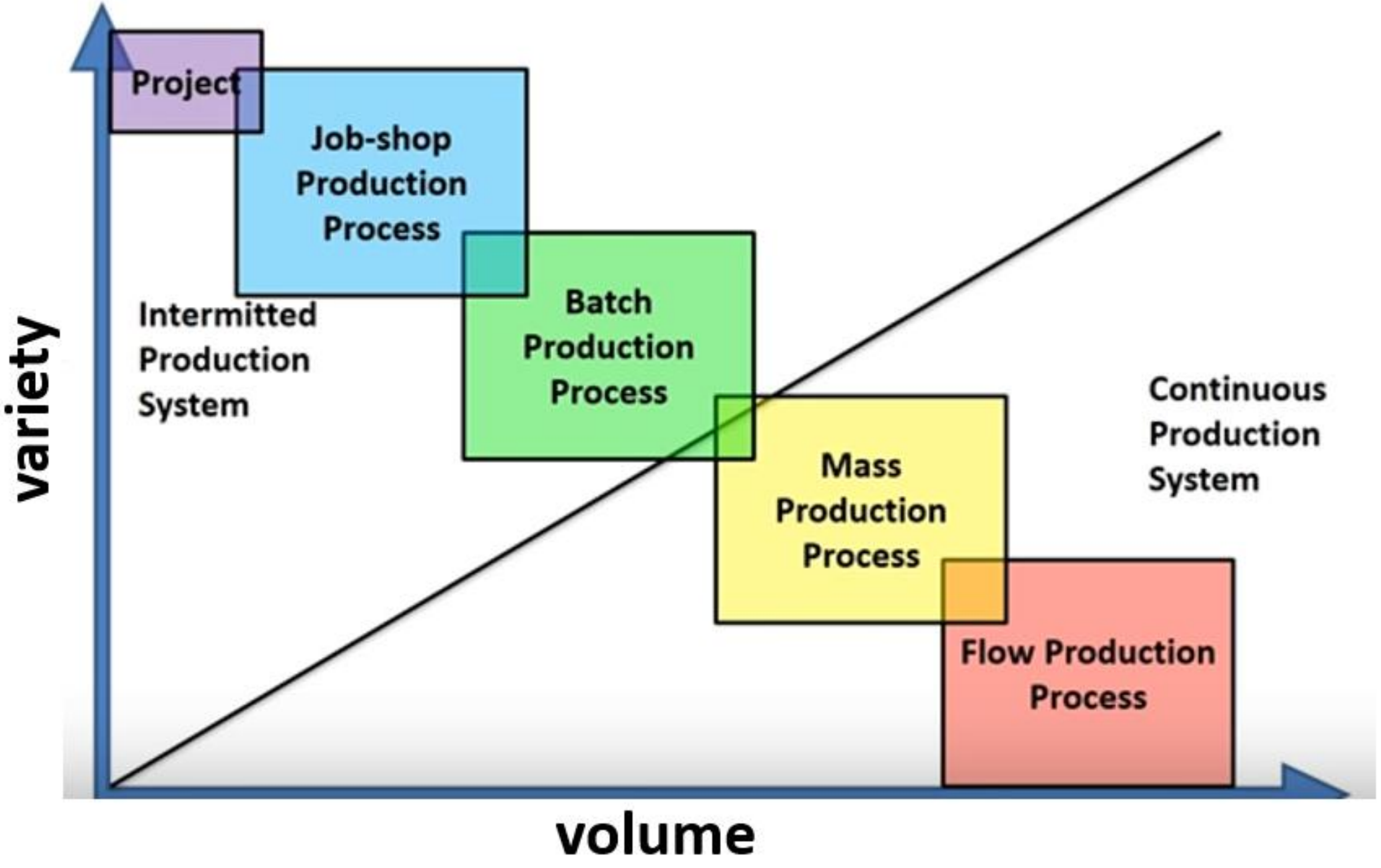
- Intermittent systems produce goods in **discrete units or batches** and are used where **product variety is high** and **volume is low to moderate**. Equipment is typically flexible, and the routing of parts can vary.

2. Continuous Production (also called repetitive or flow production)

- Continuous systems are designed for **high volume**, **low variety**, and **steady demand**. Processes are highly standardized and often highly automated.

Types of Manufacturing Systems





Project
Job-shop Production Process
Batch Production Process
Mass Production Process
Flow Production Process
Intermitted Production System
Continuous Production System
variety
volume

Project production

- A single, large, complex product is produced at a fixed location; the product doesn’t move through the factory — resources come to it.
- **Characteristics:** One-off or unique, very long duration, high customization, many distinct tasks, complex coordination.
- **Examples: Construction of bridges, buildings, ships, and custom large machinery, as well as specialized R&D prototypes.**
- **Layout:** Fixed-position layout (people, materials, equipment brought to site).
- **Challenges include scheduling numerous** interdependent activities, resource leveling, high uncertainty, and stringent quality and safety requirements.
- **Advantages:** Maximum customization; suitable for very large/unique outputs.
- **Operations focus:** Project management, planning & control, procurement scheduling, risk, and contract management.

Job production

- Small quantities of customized products; each job may have a unique routing.
- **Characteristics:** High product variety, skilled labor, general-purpose machines, intermittent workflow.
- **Examples: Machine shops making custom parts, custom furniture shops, specialist repair shops, tool rooms.**
- **Layout:** Process (functional) layout where similar machines are grouped (e.g., all lathes together).
- **Challenges:** Complex scheduling and routing, higher unit cost, variable lead times.
- **Advantages:** Highly flexible, can produce customized products quickly without retooling large lines.
- **Operations focus:** Flexible workforce, routing and scheduling (priority rules), tracking WIP, quality inspection.

Batch production

- Produces moderate quantities in batches; each batch follows the same sequence but machines are set up between runs.
- **Characteristics:** Medium volume and variety, changeover/setup times are important, economies of scale per batch.
- **Examples: Bakeries, garment factories producing seasonal clothing batches, furniture lines producing batches of the same model.**
- **Layout:** Can be either process layout or hybrid; equipment grouped but sometimes arranged to serve a set of products.
- **Challenges:** Minimizing setup times, determining optimal batch size (trade-off holding vs setup cost), scheduling batches to meet demand.
- **Advantages:** Good balance between flexibility and efficiency.
- **Operations focus:** Setup reduction (SMED), lot sizing methods (EOQ, lot-for-lot), production scheduling, inventory control.

Mass production

- A line or series of operations arranged so every unit follows the same fixed path; high volumes of standardized products.
- **Characteristics:** Very low variety, very high volumes, specialized equipment, high capital investment, short cycle time per unit.
- **Examples: Automobile assembly lines, TV or appliance assembly, consumer electronics mass lines**.
- **Layout:** Product (line) layout — machines/equipment arranged in the order of operations.
- **Advantages:** Very low unit cost, high throughput, simple scheduling, consistent quality when controlled well.
- **Challenges:** Large capital outlay, low flexibility to product change, breakdowns can stop the entire line, high dependency on supply chain reliability.
- **Operations focus:** Line balancing, takt time, preventive maintenance, quality control, inventory control (JIT/kanban often used).

Process / Continuous flow production

- The extreme form of continuous production where materials flow without discrete units; often chemical/continuous transformations.
- **Characteristics:** Continuous, 24/7 operations; output indistinguishable as separate units; highly automated and instrumented.
- **Examples: Oil refineries, chemical plants, paper, cement, power generation, sugar.**
- **Layout:** Continuous-flow process; heavily instrumented piping and process equipment.
- **Advantages:** Very efficient for commodities, low variable cost per unit, stable operations once tuned.
- **Challenges:** Very high capital and safety/regulatory requirements, complex process control, shutdowns are expensive, long lead times for changes.
- **Operations focus:** Process control, reliability engineering, safety, environmental compliance, maintenance planning, long-range throughput optimization.

Continuous Production	Intermittent Production
Unskilled Labor	Highly Skilled Labor
Investment in inventory is high	Need for inventory is minimized
Material Handling cost is less	Material handling cost is high
Permanent machine setup	Frequent changes in machine setup
Standard Products in Large quantities	Wide range of products in small quantities
Output based on anticipation of demand	Output according to order received.

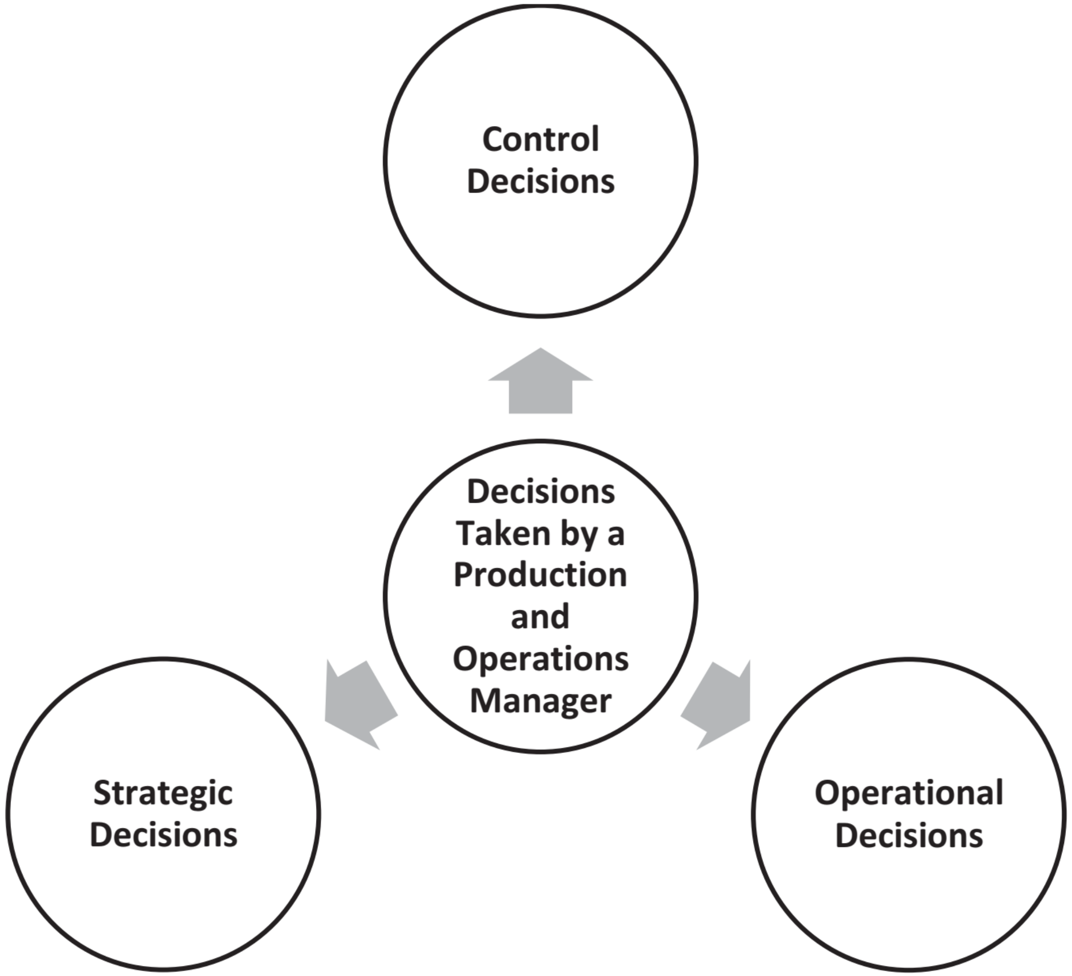
Data	*Types of Production*			
	Job Shop	***Batch***	***Repetitive or Assembly***	***Continuous***
Description of outputs	Customised goods or services	Semi standardised goods or services	Standardised goods or services	Highly standardised goods or services
Examples of goods	Speciality tools	Cookies	Automobiles	Steel, paper, flour, sugar
Examples of services	Hair Styling	Education	Car wash	Heating, air-conditioning
Examples of manufacturing facilities	Machine shop	Bakery	Assembly line	Steel mill, paper mill
Examples of service facilities	Beauty shop Barber shop	Classroom	Cafeteria lines	Central heating system
Volume	Low	Low to moderate	High	Vary high
Output variety	Very high	Moderate	Low	Very low
Equipment flexibility	Very high	Moderate	Low	Very low
Advantage	Able to handle a wide variety of work	Flexibility	Low unit cost, high volume, efficient	Very efficient very high volume
Disadvantages	Slow, high cost per unit, complex planning and scheduling	Moderate cost per unit, moderate scheduling complexity	Low flexibility, high cost of downtime	Very rigid, lack of variety, costly to change, very high cost of downtime

Decisions of a Production and Operations Manager

- The decisions made by a **Production and Operations Manager** can be broadly classified into **Strategic**, **Operational**, and **Control** decisions, as shown in the *Figure*. Each category of decision plays a vital role in ensuring efficient and effective operations.

- **1. Strategic Decisions**
- Strategic decisions are **long-term decisions** that shape the overall direction and capabilities of the production system. These decisions are typically made by top management and have a lasting impact on an organization's performance.
- They include:
  - Developing **long-term production plans**
  - Selecting, designing, and maintaining **production facilities**
  - Choosing appropriate **production technologies**
  - Designing **facility layout**
  - Making **optimal allocation of resources** (men, machines, materials, and money.

Decisions that are made by a production and operations manager



- **2. Operational Decisions:** Operational decisions are **short-term and day-to-day decisions** concerned with the smooth functioning of the production system. These decisions are generally taken by **middle- and lower-level managers**.
- They include:
  - **Scheduling** production activities
  - Deciding **what to produce and when to produce**
  - Planning and controlling **inventory of raw materials and finished goods**
  - **Materials management** and procurement planning
  - Deciding **capacity requirements** based on demand
  - **Shop floor planning and control**
- **3. Control Decisions:** Control decisions ensure that actual performance conforms to planned performance. These decisions focus on **monitoring, evaluating, and taking corrective actions**.
- They include:
  - **Controlling the quality** of products and services (Total Quality Control)
  - **Maintenance management** of machines and plants
  - **Project planning and control**
  - Monitoring productivity, costs, and delivery schedules
  - Taking corrective actions to eliminate deviations
- **Conclusion:** Effective Production and operations management requires a balanced approach to **strategic, operational, and control decisions**. Together, these decisions enable organizations to achieve efficiency, quality, flexibility, and competitiveness in both the manufacturing and service sectors.

ROLES AND RESPONSIBILITIES OF OPERATIONS MANAGER

- In today's **highly competitive business environment**, the primary objective of every organization is to produce **goods and services that meet customer preferences and ensure customer satisfaction**. To achieve this objective, the role of a **Production and Operations Manager** is of utmost importance.
- In an organization, the production and operations manager is responsible for **planning, organizing, directing, and controlling** all activities involved in **transforming inputs (men, materials, machines, methods, and money) into desired outputs**.
  - The manager must also **procure resources at minimum cost**, ensure **optimal utilization of resources**, and deliver the **right product at the right time**.

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- Producing products and services as per the desired specifications
- Acquiring raw materials at minimum prices
- Selecting the best location for facilities such as factories, warehouses, and stores
- Purchasing efficient production equipment
- Establishing work standards
- Selecting efficient production techniques
- Planning capacities of plant and machinery
- Planning and controlling production activities
- Managing and controlling inventory

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- **Measuring and monitoring productivity**
- **Maintaining industrial relations**
- **Ensuring the health and safety of employees**
- **Planning budgets and other financial resources**
- **Participating in strategic decision-making**
- **Automating production processes**
- **Enhancing research and development efforts**
- **Ensuring the timely delivery of products and services**
- **Abiding by environmental rules and regulations**
- **Maintaining long-term relationships with suppliers and other third parties**

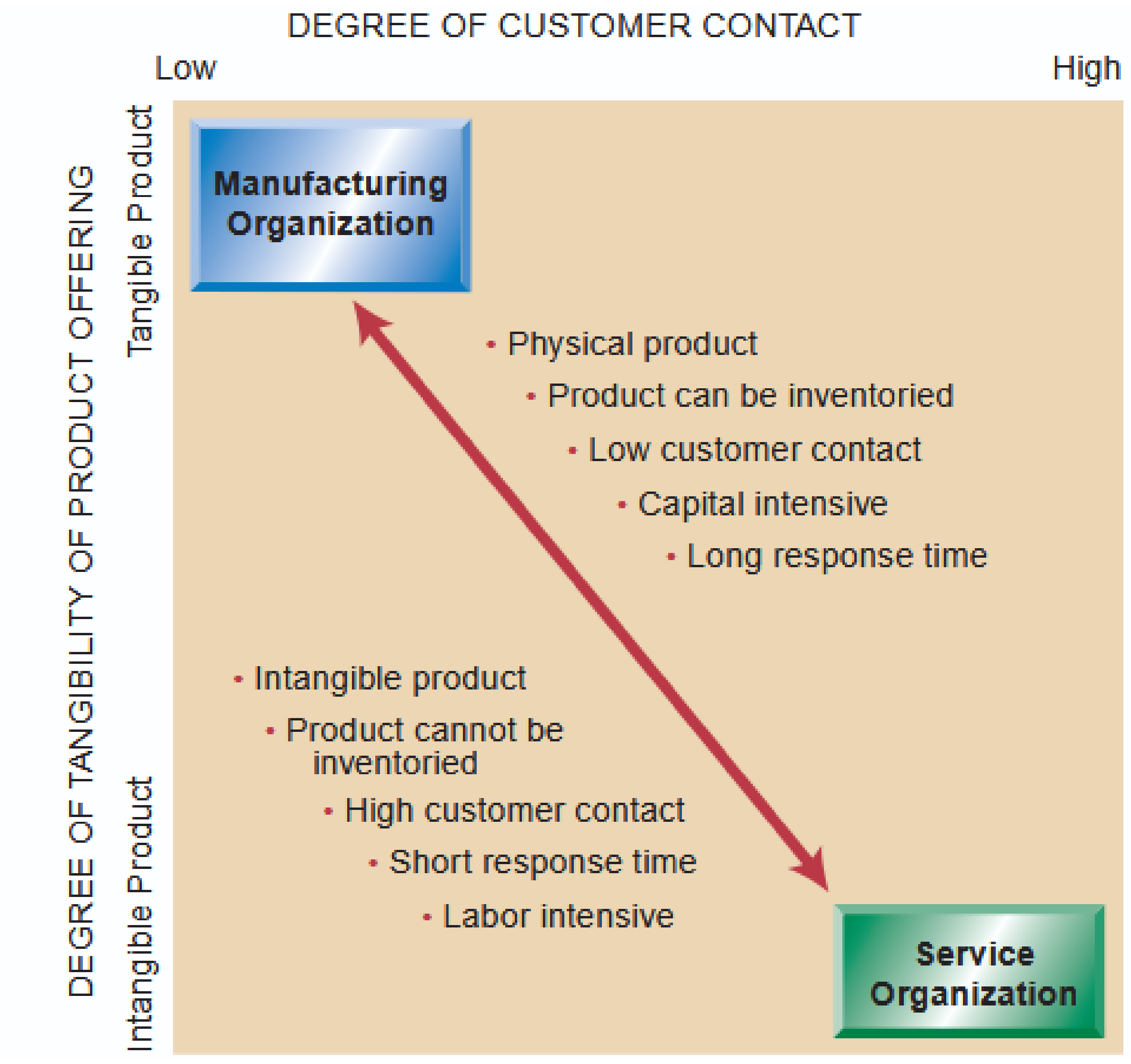
- **Product and Service Quality**
  - Producing products and services as per **desired specifications**
  - Ensuring consistency in quality and performance
- **Materials and Cost Management**
  - Acquiring **raw materials at minimum prices**
  - Maintaining long-term relationships with **suppliers and third parties**
  - Managing and controlling **inventory**
- **Facility and Equipment Planning**
  - Selecting the **best location** for factories, warehouses, and stores
  - Purchasing **efficient and appropriate production equipment**
  - Planning **capacity of plant and machinery**

- **Process and Technology Management**
  - Selecting **efficient production techniques**
  - Establishing **work standards**
  - Automating **production processes**
  - Enhancing **research and development (R&D)** efforts
- **Production Planning and Control**
  - Planning and controlling **production activities**
  - Scheduling operations and ensuring smooth workflow
  - Ensuring **timely delivery** of products and services
- **Productivity and Performance**
  - Measuring and monitoring **productivity**
  - mplementing continuous improvement practices

- **Human Resource and Industrial Relations**
  - Maintaining **industrial relations**
  - Ensuring **health and safety** of employees
  - Promoting a safe and motivating work environment
- **Financial and Strategic Responsibilities**
  - Planning **budgets and financial resources**
  - Participating in **strategic decision-making**
  - Supporting long-term organizational goals
- **Environmental and Social Responsibility**
  - Abiding by **environmental rules and regulations**
  - Promoting sustainable and eco-friendly operations

Product operations and service operations

- Operations (both manufacturing and service) involve the **conversion of inputs into outputs** through a **conversion (transformation) process**.
- During this process, **value is added**, meaning the **output has more value than the inputs** used.



The difference between Manufacturing or services

Characteristic	Manufacturing	Services
Nature of Output	Physical or Tangible	Abstract or Intangible
Degree of contact with customers	Less	Frequent
Work Process	Capital Intensive	Labour Intensive
Output Variety	Uniform	Not Uniform
Quality Assurance Issues	Standardized	More customer-centric
Performance Measurement	Standardized	More subjective
Degree of customization	Less	More

Characteristics	*Goods*	*Services*
Tangibility	Goods are tangible	Services are generally intangible
Storage/ consumption	Goods can be stored but some may not due to perishability	Services are usually consumed immediately
Nature	Products may be of different varieties with respect to quality or size but generally do not depend on customers	Each service to different customers may be unique as per the requirement of customers
Quality parameters	Quality parameters of goods are measurable	Quality parameters in production of services are subjective and generally depend on the perception of the customer.
Importance of location	Location in production of goods may be important only due to freight factor	Location in the case of services is very important because of physical presence of the customer at the site of services
Involvement of technology, knowledge and skills	Production of goods is necessarily technology based	Services operations are more or less knowledge-and skill-based, but many times technology is also required to produce services
Involvement of customer	Production of goods does not require customer interaction at the facility	In the case of services, high degree of customer contact facilitates customized service quality
Transferability	Most of the goods are transferable	Services may not be transferable, and are usually consumed immediately

Characteristics	*Production Management*	*Operations Management*
Nature of output	Manufacturing of tangible goods, e.g. car, bike, etc.	Producing both tangible and intangible output (services), e.g. banking, transportation, etc.
Consumption of output	Tangible products are consumed over a period of time	Intangible services have to be consumed immediately
Nature of work	Manufacturing of tangible products requires less labour and more equipment	Producing the intangible services needs more labour and less equipment
Customer participation in conversion	No participation	Frequent participation while providing the services

Point of Comparison	Manufacturing Operations	Service Operations
Degree of Customer Contact	Low to moderate; customers rarely interact directly with production	High, especially in services like retail, hospitality, and healthcare
Labor Content of Jobs	Often lower, with a higher degree of automation	Often higher, although some automated services exist
Uniformity of Inputs	Higher, more controlled and consistent inputs	Lower, more variability and uniqueness in each service instance
Measurement of Productivity	Easier to measure due to consistent outputs and inputs	More difficult due to variability and complexity of inputs

Quality Assurance	Easier to control quality, mistakes can be corrected before delivery	More challenging, with less opportunity to correct mistakes before customer exposure
Inventory	High use of inventory, products can be stored	Low use of inventory, services cannot be stored and must be provided on demand
Wages	Often higher and more uniform	Varies widely from high-paid professionals to minimum-wage workers
Ability to Patent	Easier to patent product designs	Harder to patent service designs, making them easier to copy

Differences Between Service and Manufacturing Organizations

Tangibility of Output

- **Service Organizations**: Provide intangible outputs like consultancy, training, or maintenance.
- **Manufacturing Organizations**: Produce tangible goods that customers can see and touch.

Production on Demand

- **Service Organizations**: Do not hold inventory; services are created as needed by the client.
- **Manufacturing Organizations**: Produce goods for inventory based on demand forecasts. Some manufacturers maintain minimum stock levels and rely on just-in-time production to meet demand.

Customer-Specific Production

- **Service Organizations**: Tailor services to individual customer needs, although they design the scope and content in advance.
- **Manufacturing organizations can produce goods without a specific customer order; however,** it is risky to produce items that do not meet market needs.

Labor Requirements and Automated Processes

- **Service Organizations**: Labor-intensive, requiring skilled personnel for service delivery. Automation is limited, though knowledge management systems help.
- **Manufacturing Organizations**: Can automate many processes to reduce labor requirements, although some manufacturing remains labor-intensive, especially where labor costs are low.

Physical Production Location

- **Service Organizations**: Do not need a physical production site; services can be delivered from anywhere using communication networks.
- **Manufacturing Organizations**: Require physical locations for production and inventory. Production can occur at various points in the supply chain, not necessarily at the manufacturer's site.

Manufacturing Operations

- Manufacturing operations are the activities through which **raw materials are transformed into finished, tangible products** using **manpower, machines, money, management, and methods**.
- The main objective is **value addition**, i.e., the finished product has **greater value than the inputs**.
- Manufacturing operations are **fundamental processes that can be classified based on the changes in shape, size, or physical/chemical properties** of materials.

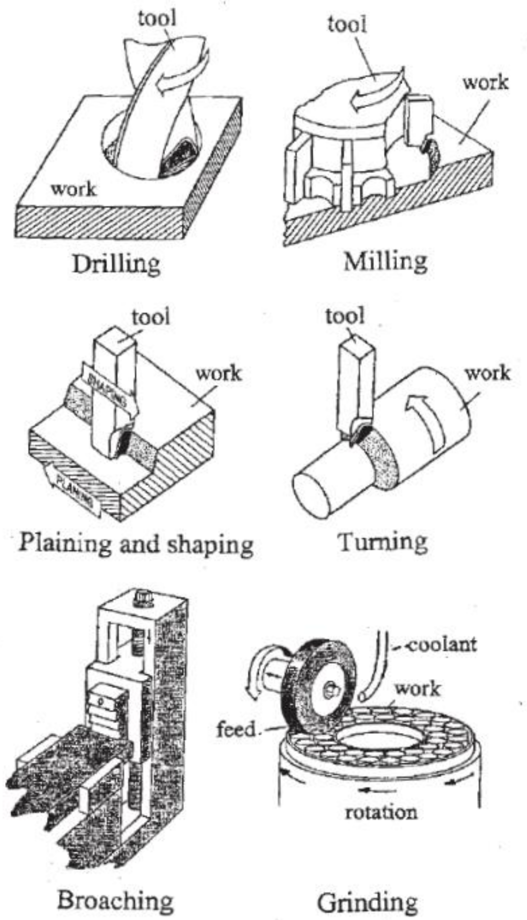
- **Classification of Manufacturing Operations**
  - **Forming processes**
  - **Machining processes**
  - **Assembly processes**
  - **Treatment processes**
  - **Fabrication processes**
  - **Chemical processes**
  - **Testing processes**

Forming Processes

- Forming processes **change the physical shape and size of a material without removing material**. These processes are widely used in the metal and plastic industries to produce components of desired geometry.
- **Types of Forming Processes**
  - **1. Casting:** Casting is a process in which **molten metal is poured into a mould cavity** and allowed to solidify to obtain the desired shape.
    Examples: Sand casting, die casting.
- **2. Forging:** Forging is a process in which **heated metal is shaped by compressive force using a die**. The metal takes the shape of the die.
  Types: Hot forging and cold forging.
- **3. Moulding:** Moulding is a **manufacturing process for plastics** (thermoplastic and thermosetting). Plastic material is heated, mixed, and **forced into a mould cavity**, where it cools and solidifies.
- **4. Extrusion:** Extrusion is a process used to produce **long components with a uniform cross-section**. The material is **forced through a die** to obtain the desired shape.
  Types: Forward extrusion, backward extrusion, and impact extrusion.

- **5. Stamping:** Stamping is a **cold working process** in which force is applied using a **die and punch** to cut or shape sheet metal.
- **6. Embossing:** Embossing is a process in which **designs, letters, or patterns are raised or depressed** on a sheet metal surface using dies.
- **7. Spinning:** Spinning is a process of shaping a **metal sheet over a rotating mandrel** on a high-speed lathe to form hollow, axisymmetric parts
- **8. Coining:** Coining is a **compressive forming process** in which shallow impressions or fine details are produced on the surface of a blank using high pressure.

- **Machining Processes**
- Machining processes are **material removal processes** in which excess material is removed from a workpiece in the form of chips to obtain an **accurate shape, size, and surface finish**.
- **Common Machining Processes**
  - Turning
  - Milling
  - Drilling
  - Grinding
  - Shaping and planning
- **Purpose**
  - To achieve **dimensional accuracy**
  - To obtain a **smooth surface finish**
  - To produce complex geometries.



- **Fabrication Processes**
- Fabrication is a manufacturing process that involves **cutting, shaping, and joining metal components**, usually made from **metal plates, sheets, or structural sections**, to produce a finished product or structure.
  It mainly involves **cutting, bending, welding, forging, and assembly operations**.
- Fabrication is commonly used in **construction, automotive, shipbuilding, railway, and heavy engineering industries**.
- **Types of Fabrication**
- **Light Fabrication**
  Uses thin sheets; products are small and light.
  Examples: Furniture, electrical panels.
- **Medium Fabrication**
  Uses medium-thickness plates; products are moderate in size.
  Examples: Storage tanks, automobile frames.
- **Heavy Fabrication**
  Uses thick plates and heavy sections; products are very large and heavy.
  Examples: **Locomotives, earth-moving equipment**.

- **Assembly Processes**
- Assembly processes involve **joining two or more parts** to form a complete product or sub-assembly.
- **Types of Assembly Processes**
- **Permanent joining: Welding, brazing, soldering, riveting**
- **Temporary joining: Nuts, bolts, screws, pins**
- **Purpose**
- To form a **finished product**
- To ensure **proper functioning and the strength** of components

- **Treatment Processes**
- Treatment processes are used to **improve the mechanical or physical properties** of materials without changing their shape.
- **Types of Treatment Processes**
- Heat treatment (annealing, normalizing, hardening, tempering)
- Surface treatment (carburizing, nitriding, electroplating)
- **Purpose**
- To improve **strength, hardness, ductility, and wear resistance**
- To enhance **service life** of components
-

- **1. Heat Treatment Processes**
- Heat treatment involves **controlled heating and cooling of metals** to obtain desired properties.
- **Annealing**
  - A **softening process**.
  - Metal is heated and then **slowly cooled inside the furnace**.
  - Removes internal stresses and improves **machinability and ductility**.
- **Normalizing**
  - Metal is heated above the critical temperature and then **cooled in still air outside the furnace**.
  - Improves **strength and grain structure**.
- **Hardening**
  - Metal is heated to a high temperature and then **rapidly cooled (quenched)** in **water, oil, or brine**.
  - Increases **hardness and wear resistance**.
- **Tempering**
  - Hardened metal is **reheated below the critical temperature**, held for some time, and then cooled.
  - Reduces brittleness and improves **toughness**.

- **2. Surface Treatment Processes**
  - Surface treatment is applied to metals and alloys for **industrial use, corrosion protection, and improved appearance**.
- **Purpose**
  - Protection from the atmosphere
  - Improved surface finish
  - Increased corrosion and wear resistance
- **Common Surface Treatment Methods**
  - Electroplating
  - Powder coating
  - Galvanizing
  - Anodizing
  - Metal spraying
  - Phosphating
  - Hot dipping
  - Acid and alkaline treatment
  - Vapour degreasing
  - Ultrasonic cleanin

- **Testing Processes:** Testing processes are **distinct from manufacturing processes, yet they play a crucial role in ensuring the quality, reliability, and performance** of products. Testing helps identify defects and verify whether a product meets the required standards.
- Testing processes are broadly classified into **two types**:

- **1. Destructive Testing (DT):** In destructive testing, the **test specimen is damaged or destroyed** during testing to determine its mechanical properties.
- **Purpose**
  - To find strength limits
  - To identify internal and external defects
- **Examples**
  - Tensile test, Compression test, Impact test, Hardness test, Fatigue test
- **2. Non-Destructive Testing (NDT):** In non-destructive testing, the **product is tested without causing any damage**, and it can be used after testing.
- **Purpose**
- To detect surface and internal defects
- To ensure product safety and reliability
- **Examples**
- Ultrasonic testing, Radiographic testing (X-ray), Magnetic particle testing, Dye penetrant testing, Eddy current testing etc.

Service Operations

- **Service operations** refer to the activities involved in producing and delivering intangible outputs, specifically services. Unlike manufacturing, services do not result in physical goods.
- Examples of service operations include:
  - Education
  - Healthcare
  - Transportation
  - Hospitality
  - Banking and insurance
  - Entertainment
- In developing countries such as India and China, service operations play a crucial role in driving economic growth and contribute substantially to the **Gross Domestic Product (GDP)**. After **liberalization, privatization, and globalization**, the service sector has become a major contributor to national economies.

Basis of Classification	Type of Service Operation	Description	Examples
Customer Contact	High-contact services	Direct and continuous interaction with customers	Hospitals, Hotels, Education
	Low-contact services	Minimal or no direct customer interaction	Insurance processing, Back-office banking
Degree of Customization	Standardized services	Same service provided to all customers	ATM services, Public transport
	Customized services	Services tailored to individual needs	Medical treatment, Legal services
Nature of Service	People-processing services	Services performed on people	Airlines, Hospitals
	Possession-processing services	Services performed on customer-owned goods	Vehicle servicing, Equipment repair
	Information-processing services	Services related to data and information	Banks, IT services
Service Delivery Process	Professional services	Low volume, high customization	Doctors, Consultants
	Service shop	Medium volume, medium customization	Hospitals, Repair centers
	Service factory	High volume, low customization	Fast food chains, Airlines
Tangibility	Pure services	Completely intangible services	Education, Counseling
	Services with supporting goods	Services supported by tangible products	Restaurants, Retail stores

Various Measures of Capacity

Business	Inputs	Outputs
Auto Manufacturing	Labor hours, machine hours	Number of cars per shift
Steel Mill	Furnace size	Tons of Steel per day
Oil Refinery	Number of acres,	Gallons of fuel per day
Farming	Number of cows	Bushels of grain per acre per year
Restaurant	Number of tables, seating capacity	Number of meals served per day
Theater	Number of seats	Number of tickets sold per performance
Retail sales	Square feet of floor space	Revenue generated per day

Product Operations Examples

Automobile Manufacturing

- **Output**: Cars and trucks.
- **Processes**: Assembly line production, quality control, painting, and finishing.
- **Key Resources**: Raw materials (steel, plastic, rubber), machinery, labor, and assembly plants.

Electronics Manufacturing

- **Output**: Smartphones, laptops, and televisions.
- **Processes**: Circuit board assembly, software installation, product testing, and packaging.
- **Key Resources**: Electronic components, skilled labor, automated machinery, and clean rooms.

Food Production

- **Output**: Packaged foods, beverages, and processed meats.
- **Processes**: Ingredient mixing, cooking, packaging, and labeling.
- **Key Resources**: Raw ingredients, processing equipment, quality control labs, and cold storage facilities.

Service Operations Examples

Consulting Services

- **Output**: Business strategy reports, market analysis, and implementation plans.
- **Processes**: Client meetings, data analysis, report preparation, and presentations.
- **Key Resources**: Knowledgeable consultants, analytical tools, office space, and communication technology.

Healthcare Services

- **Output**: Medical diagnoses, treatments, and surgeries.
- **Processes**: Patient consultations, diagnostic tests, medical procedures, and follow-up care.
- **Key Resources**: Medical professionals, diagnostic equipment, treatment facilities, and pharmaceuticals.

Education Services

- **Output**: Educational courses, degrees, and certifications.
- **Processes**: Curriculum development, classroom instruction, assessments, and student support services.
- **Key Resources**: Educators, learning materials, classrooms, and online learning platforms.

Functions of Production and Operations Management

- Production and Operations Management (POM) involves **planning, organizing, directing, and controlling** activities related to the conversion of inputs into outputs. Control and decision-making powers are exercised at **different levels of management**, and in many organizations, these functions are **handled by a single operations manager**.

- The **common functions of the operations department** are explained below:
- **1. Selection of Material**
  - The operations manager selects the **appropriate raw materials** required for the product. This involves **researching product requirements, quality, cost, and the availability** of materials.
- **2. Selection of Appropriate Method**
  - Choosing a suitable **method of production** is a crucial function. The method is selected based on **available resources, technology, demand, and constraints**.
- **3. Selection of Suitable Machinery and Equipment**
  - The operations department selects **machinery and equipment** according to the **product design, production volume, and process requirements**.

- **4. Fixation of Production Targets and Delivery Dates:** Production targets and delivery schedules are fixed to **meet customer demand** while **minimizing production cost** and ensuring timely delivery.

- **5. Cost Control:** To survive in a competitive environment, the operations department ensures that **production costs are maintained at the desired level** through efficient use of resources.

- **6. Scheduling of Activities:** Scheduling involves **planning and allocating work** to manpower, machines, and departments so that production is completed **on time**.

- **7. Routing and Layout of Production Lines:** Routing determines the **path followed by raw materials and components** through various operations. A proper layout ensures a smooth flow of materials and minimizes **handling**.

- **8. Dispatching:** Dispatching involves **issuing work orders and documents**, such as routing sheets and inspection cards, to begin production activities.

- **9. Follow-up (Expediting):** Follow-up ensures that **scheduled activities are carried out as planned**. Actual progress is compared with the plan, and **corrective actions** are taken if deviations occur.

- **10. Inspection:** Inspection is carried out to **maintain quality standards**. Products and components are checked to ensure they **conform to specifications**.

Current Trends in Operations Management

- Production and operations management has undergone significant changes due to **market dynamics, globalization, technology, and customer expectations**. The following are some important recent trends.

- **1. Change of Focus – From Seller to Buyer/Customer**
  - Traditionally, production systems were **seller-oriented**, where goods were produced first and then sold in the market. Products were treated as **commodities**, and the focus was on mass production.
  - In the present scenario, production systems are **buyer or customer-oriented**. Products are designed and produced based on **customer needs and preferences**. Companies differentiate products to **target specific market segments or niches**.

- **2. Globalization**
  - Globalization has brought major changes in production and operations, such as:
  - Access to **global markets and customers**
  - Opportunities for **product variety and customization**
  - **Horizontal and vertical integration** to control supply chains and expand business
  - Increased competition from **unexpected global players**
  - Outsourcing and offshoring of operations

- **3. Production System: Contribution of Japan**
  - Japanese production systems have had a significant influence on modern operations management. Important contributions include:
  - **Total Quality Management (TQM)**
  - **Just-in-Time (JIT)** production
  - **Continuous improvement (Kaizen)**
  - **Designing products based on customer requirements**
  - The focus has shifted from improving past methods to **rethinking and optimizing business processes**.

- **4. Growth of Service Operations**
  - Since the 1960s, there has been a steady **increase in service employment**, and today **service jobs outnumber manufacturing jobs**. Operations management principles are now widely applied in **banks, hospitals, education, transportation, and hospitality sectors**.

5. Environmental and Social Issues

- There is increasing awareness about **environmental protection and social responsibility**. Governments and global organizations have introduced regulations and standards such as:
- **Environmental management systems**
- **ISO 14000 standards** for environmental performance
- Organizations are expected to reduce pollution and conserve resources.

- **6. Ethical Production**
  - Ethical production emphasizes **minimizing environmental impact** and responsible use of resources. Product design now considers:
  - Safe manufacturing practices
  - Recycling and reuse
  - **Disposal of products at the end of their life cycle**

- **7. New Technologies**
  - Advances in technology have transformed production and operations. Major developments include:
  - Robotics and automation
  - Computer-controlled manufacturing
  - Biotechnology
  - Global Positioning System (GPS)
  - Information and communication technologies (ICT)
  - These technologies have led to **higher productivity, accuracy, and efficiency**.

- **Customer Relationship Management (CRM):** CRM is a strategy for managing customer interactions to enhance satisfaction and loyalty.
- **key points:**
  - Helps organizations better understand, acquire, and retain customers.
  - Involves using technologies such as databases, analytics, and communication tools.
  - Aims to improve **customer service**, enhance customer experience, and increase loyalty.
  - Leads to higher customer retention and long-term profitability.

- **Business Process Reengineering (BPR):** BPR focuses on the radical redesign of business processes to achieve major improvements in performance.
- **Key points:**
  - Involves rethinking current workflows, eliminating unnecessary steps, and simplifying processes.
  - Helps an organization identify weaknesses or bottlenecks in existing systems.
  - Results in dramatic improvements in **efficiency, speed, and cost reduction**.
  - Often supported by information technology to streamline operations.

- **Environmental Issues and Sustainability:** Environmental awareness has become essential for modern manufacturing organizations.
- **Key points:**
  - Organizations aim to reduce pollution, emissions, and waste.
  - Emphasis on **recycling**, **reuse**, and reducing carbon footprint.
  - Use of **biodegradable packaging** and **eco-friendly materials** is increasing.
  - Green manufacturing and compliance with environmental regulations (such as ISO 14001) are important.
  - Sustainable production enhances corporate image and long-term stability.

- **8. Flexibility**
  - Flexibility refers to the ability to **quickly adjust production levels or shift production capacity from one product or service to another in response to** market demand.

- **9. Time Reduction**
  - Reducing **manufacturing cycle time and delivery time** has become a major competitive advantage. Companies focus on:
  - Faster processing
  - Reduced lead time
  - Quick response to customer demand

- **Conclusion**
  - Recent trends in production and operations emphasize **customer focus, quality, sustainability, technology, flexibility, and speed**, enabling organizations to remain competitive in a dynamic global environment.

Sample Questions

- **Introduction to Operations Management**
- **Questions (Short)**
  - Define Operations Management.
  - State any two objectives of operations management.
  - Mention two examples of operations in service and manufacturing organizations.
  - What are operations?
  - Differentiate between production and operations.
  - What are inputs and outputs in an operating system?
  - Why is Operations Management called a core function of an organisation?
- **Questions (Long)**
  - Explain the concept of Operations Management.
  - Explain the scope of operations management.
  - Discuss the importance of operations management in an organization.
  - Explain the transformation process in operations management.
  - Explain Operations Management in detail and discuss its role in achieving organizational objectives.
  - Explain the functions of operations management with suitable examples.
  - Discuss how operations management contributes to competitive advantage.
  - Explain the relationship between operations management and other functional areas of management.

Types of Manufacturing Systems

Questions (Short)

- What is a manufacturing system?
- Define job production.
- What is batch production?
- State one advantage of mass production.
- Why is flexible manufacturing preferred in modern industries?

Questions (Long)

- Explain the job production system with an example.
- Discuss batch production and its suitability for small-scale industries.
- Explain the mass production system and its characteristics.
- Compare batch production and mass production systems.
- Describe various types of manufacturing systems with examples.
- Explain the continuous production system and its applications.
- Analyze the selection criteria for choosing a suitable manufacturing system.
- Compare job, batch, and mass production systems from an operational perspective.

- History of Operations Management

- **Questions (Easy)**
  - Who is known as the father of Scientific Management?
  - Define Scientific Management.
  - What is mass production?
  - What is the Hawthorne experiment related to?
  - Name any two contributors to operations management.
- **Questions (Long)**
  - Trace the historical development of operations management from the Industrial Revolution to the present day.
  - Explain Scientific Management principles and their relevance today.
  - Discuss the major milestones in the evolution of operations management.
  - Critically analyze the contributions of Taylor, Ford, and Elton Mayo to operations management.

- Roles and Responsibilities of Operations Manager
- **Questions (Easy)**
  - Who is an operations manager?
  - State any two roles and responsibilities of an operations manager.
- **Questions (Long)**
  - Explain the roles and responsibilities of an operations manager in detail.
  - Discuss the strategic and operational roles of an operations manager.
  - Explain how an operations manager contributes to productivity improvement.
  - Describe the role of the operations manager in managing resources effectively.
  - Explain the challenges faced by operations managers in modern organizations.

- **Product Operations and Service Operations**
- **Questions (Easy)**
  - Define product operations.
  - Define service operations.
  - Give two examples of service operations.
  - Mention one difference between product and service.
- **Questions (Long)**
  - Explain product operations with examples.
  - Describe service operations and their characteristics.
  - Differentiate between product and service operations.
  - Explain the concept of customer involvement in service operations.
  - Discuss challenges in managing service operations.
  - Explain product operations and service operations in detail with suitable examples.
  - Compare product-based and service-based operations.
  - Discuss the unique characteristics of service operations and their impact on operations management.
  - Explain the role of operations management in service industries.
  - Describe how technology influences product and service operations.

- **Current Trends in Operations Management**
  - Explain the current trends in operations management in detail.
  - What is meant by the shift from seller-oriented to buyer-oriented production?
  - List any two recent trends in production & operations management.
  - Explain the major recent trends in production and operations management.

Thankyou